PHYS 4027 / 5027 · FALL 2026 · LECTURE 15

Computing QR, and the SVD

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Last time: $\mathbf{A} = \mathbf{QR}$

- $\mathbf{Q}$ has orthonormal columns, $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$; $\mathbf{R}$ is upper triangular.
- We used it to solve systems ($\mathbf{Rx} = \mathbf{Q}^T\mathbf{b}$) and to find eigenvalues (the QR iteration). It also works for rectangular matrices.
- But we never said how to find $\mathbf{Q}$ and $\mathbf{R}$.

Today: two ways to compute QR, one of which fails in floating point. Then the decomposition that exists for every matrix: the SVD.

Part I · Gram-Schmidt

Straighten the columns one at a time

Think in columns

$$\mathbf{A} = \begin{pmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{pmatrix}$$

Goal: orthonormal vectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ (the columns of $\mathbf{Q}$), built so that

- $\mathbf{e}_1$ points along $\mathbf{a}_1$;
- $\mathbf{e}_2$ lies in the plane of $\mathbf{a}_1$ and $\mathbf{a}_2$;
- in general, each $\mathbf{a}_i$ is a combination of $\mathbf{e}_1, \ldots, \mathbf{e}_i$ only.

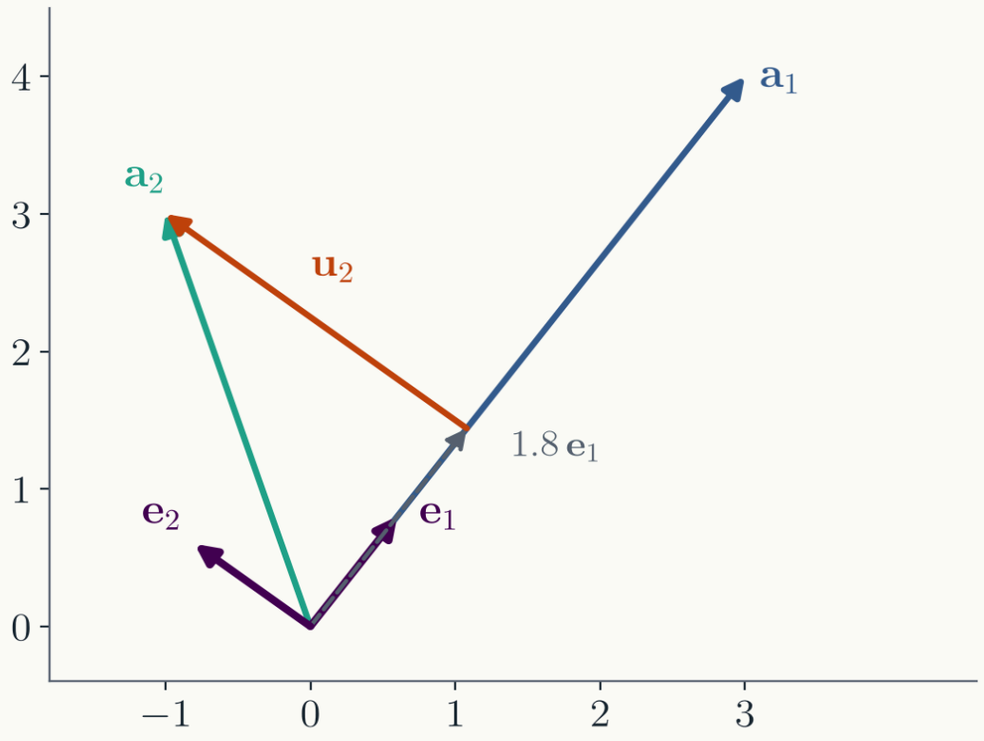
That last property is what makes $\mathbf{R}$ upper triangular.

The Gram-Schmidt process

From each column, remove its components along all the earlier $\mathbf{e}_j$, then normalize what is left:

$$
\begin{aligned}
\mathbf{u}_1 &= \mathbf{a}_1, & \mathbf{e}_1 &= \mathbf{u}_1 / |\mathbf{u}_1| \\
\mathbf{u}_2 &= \mathbf{a}_2 - (\mathbf{e}_1 \cdot \mathbf{a}_2)\,\mathbf{e}_1, & \mathbf{e}_2 &= \mathbf{u}_2 / |\mathbf{u}_2| \\
\mathbf{u}_3 &= \mathbf{a}_3 - (\mathbf{e}_1 \cdot \mathbf{a}_3)\,\mathbf{e}_1 - (\mathbf{e}_2 \cdot \mathbf{a}_3)\,\mathbf{e}_2, & \mathbf{e}_3 &= \mathbf{u}_3 / |\mathbf{u}_3| \\
&\;\;\vdots \\
\mathbf{u}_i &= \mathbf{a}_i - \sum_{j<i} (\mathbf{e}_j \cdot \mathbf{a}_i)\,\mathbf{e}_j, & \mathbf{e}_i &= \mathbf{u}_i / |\mathbf{u}_i|
\end{aligned}
$$

In two dimensions



start	$\mathbf{a}_1 = (3,\ 4), \quad \mathbf{a}_2 = (-1,\ 3)$
normalize $\mathbf{a}_1$	$\lvert\mathbf{a}_1\rvert = \sqrt{3^2 + 4^2} = 5$ $\mathbf{e}_1 = \mathbf{a}_1/5 = (0.6,\ 0.8)$
how much $\mathbf{e}_1$ is in $\mathbf{a}_2$	$\mathbf{e}_1 \cdot \mathbf{a}_2 = 0.6 \cdot (-1) + 0.8 \cdot 3 = 1.8$
remove it	$\mathbf{u}_2 = \mathbf{a}_2 - 1.8\,\mathbf{e}_1$ $= (-1,\ 3) - (1.08,\ 1.44) = (-2.08,\ 1.56)$
normalize $\mathbf{u}_2$	$\lvert\mathbf{u}_2\rvert = \sqrt{2.08^2 + 1.56^2} = 2.6$ $\mathbf{e}_2 = \mathbf{u}_2/2.6 = (-0.8,\ 0.6)$
check	$\mathbf{e}_1 \cdot \mathbf{e}_2 = -0.48 + 0.48 = 0$

Reading off $\mathbf{R}$

Turn the process around and write each column in terms of the $\mathbf{e}$'s:

$$\mathbf{a}_1 = |\mathbf{u}_1|\,\mathbf{e}_1, \qquad \mathbf{a}_2 = (\mathbf{e}_1 \cdot \mathbf{a}_2)\,\mathbf{e}_1 + |\mathbf{u}_2|\,\mathbf{e}_2, \qquad \ldots$$

In matrix form, that is $\mathbf{A} = \mathbf{Q}\mathbf{R}$, with $r_{ji} = \mathbf{e}_j \cdot \mathbf{a}_i$ above the diagonal and $r_{ii} = |\mathbf{u}_i|$ on it. For our example:

$$\begin{pmatrix} 3 & -1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} 5 & 1.8 \\ 0 & 2.6 \end{pmatrix}$$

Every entry of $\mathbf{R}$ is a number we already computed along the way.

Gram-Schmidt can fail

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ \varepsilon & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}, \qquad \varepsilon = 10^{-8}$$

Three columns, nearly parallel but independent. Run the same steps, rounding where the computer rounds.

normalize $\mathbf{a}_1$	$\lvert\mathbf{a}_1\rvert = \sqrt{1 + 10^{-16}}$, and $1 + 10^{-16}$ rounds to 1 $\mathbf{e}_1 = \mathbf{a}_1 / 1 = (1,\ \varepsilon,\ 0,\ 0)$
second column	$\mathbf{e}_1 \cdot \mathbf{a}_2 = 1 \cdot 1 + \varepsilon \cdot 0 = 1$, so $\mathbf{u}_2 = \mathbf{a}_2 - \mathbf{e}_1 = (0,\ -\varepsilon,\ \varepsilon,\ 0)$ $\lvert\mathbf{u}_2\rvert = \sqrt{2}\,\varepsilon$, so $\mathbf{e}_2 = (0,\ -1,\ 1,\ 0)/\sqrt{2}$
third column	$\mathbf{e}_1 \cdot \mathbf{a}_3 = 1$ and $\mathbf{e}_2 \cdot \mathbf{a}_3 = 0$, so $\mathbf{u}_3 = \mathbf{a}_3 - \mathbf{e}_1 = (0,\ -\varepsilon,\ 0,\ \varepsilon)$ $\mathbf{e}_3 = (0,\ -1,\ 0,\ 1)/\sqrt{2}$
check	$\mathbf{e}_2 \cdot \mathbf{e}_3 = \frac{1}{2}(0 + 1 + 0 + 0) = \frac{1}{2}$, not 0

The "orthonormal" vectors $\mathbf{e}_2$ and $\mathbf{e}_3$ are 60° apart. The computer returns exactly this.

What went wrong

- Only one rounding mattered: $1 + \varepsilon^2 \to 1$ in $|\mathbf{a}_1|$. It changed $\mathbf{e}_1$ by about $\varepsilon^2 = 10^{-16}$. Harmless, it seems.
- In exact arithmetic that correction leaks into $\mathbf{u}_2$, whose first entry is ε^2 rather than 0. After dividing by $|\mathbf{u}_2| \approx \sqrt{2}\,\varepsilon$, the first entry of $\mathbf{e}_2$ is about $\varepsilon/\sqrt{2}$, and $\mathbf{e}_2 \cdot \mathbf{a}_3 \approx \varepsilon/\sqrt{2}$ instead of 0.
- So the term $(\mathbf{e}_2 \cdot \mathbf{a}_3)\,\mathbf{e}_2$ that the rounded run skipped has size ε. And $\mathbf{u}_3$ itself has size ε. The skipped term is as big as the answer.

Subtracting nearly equal vectors leaves a small remainder, and every small error is then a large fraction of it: catastrophic cancellation (Lecture 1), one vector at a time.

How close to dependent?

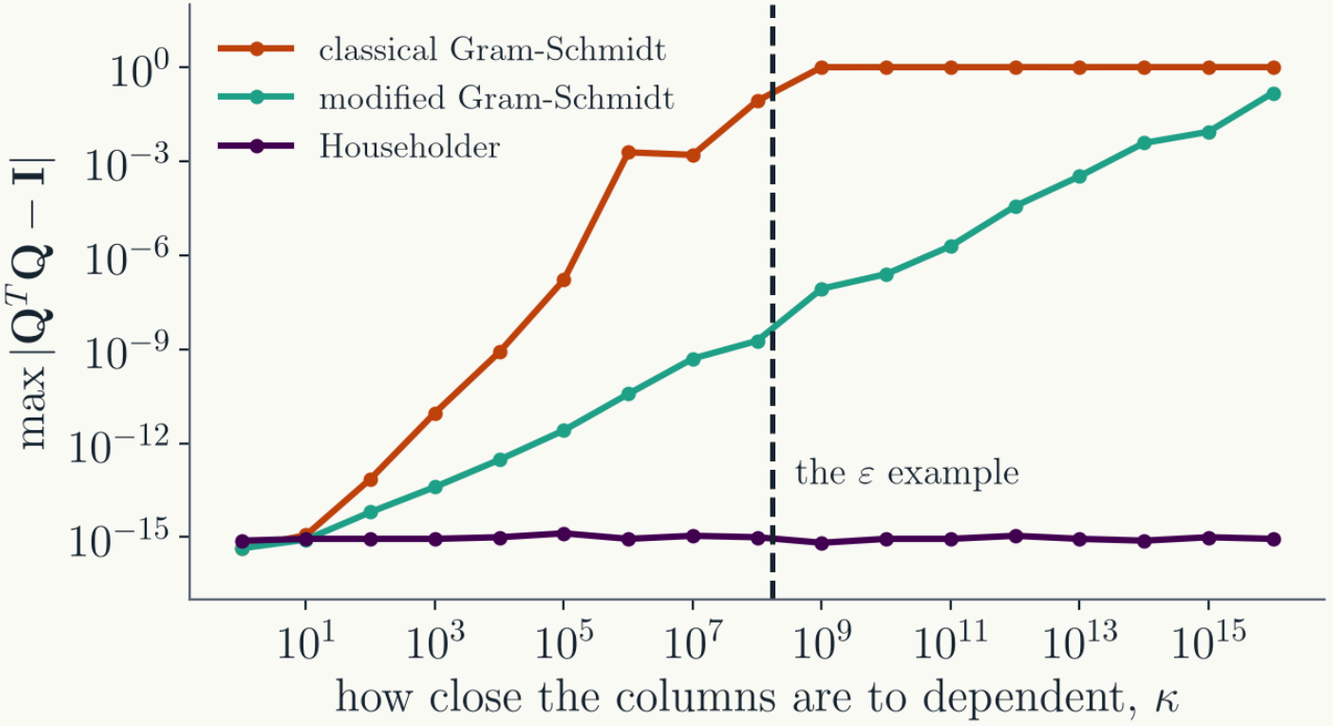
Combine the columns with unit-length coefficients: $\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,\ |\mathbf{x}| = 1.$

$$\kappa = \frac{\text{length of the longest such vector}}{\text{length of the shortest}}$$

- Orthonormal columns: every unit combination has length 1, so $\kappa = 1$.
- The ε example: $(\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3)/\sqrt{3}$ has length about $\sqrt{3}$; $(\mathbf{a}_1 - \mathbf{a}_2)/\sqrt{2}$ has length ε. So $\kappa \approx \sqrt{3}/10^{-8} \approx 1.7 \times 10^{8}$.

Large κ means some combination of the columns nearly cancels. That is exactly where Gram-Schmidt subtracts nearly equal vectors.

How bad, in general



60×30 random matrices, κ from 1 to 10^{16}. Dashed line: the ε example.

Classical Gram-Schmidt (the process above): complete loss by $\kappa \sim 10^{9}$.

Modified Gram-Schmidt, the same subtractions in a different order: error grows in proportion to κ.

Householder (next part): rounding level, always.

Part II · Householder reflections

Reflect instead of subtract

The goal

Find an orthogonal matrix $\mathbf{P}$ that turns a column into a multiple of $\mathbf{e}_1$:

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \quad \longrightarrow \quad \mathbf{P}\mathbf{a} = \begin{pmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

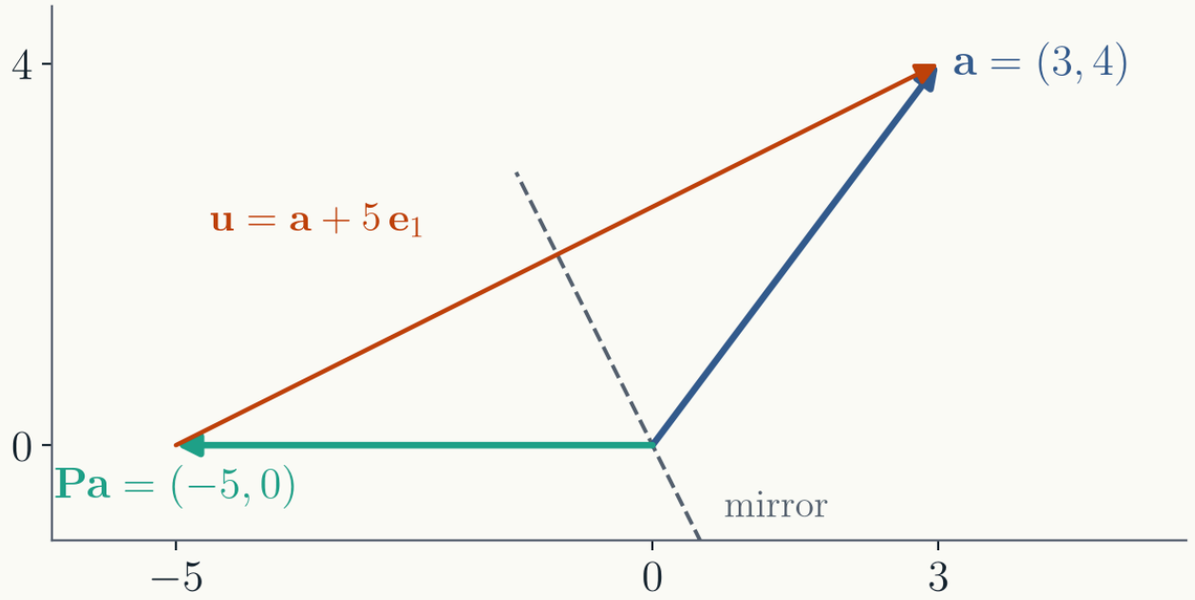
- An orthogonal matrix keeps lengths, so $\alpha = \pm|\mathbf{a}|$.
- Applied to the first column of $\mathbf{A}$, this zeros everything below the diagonal, in one step. No subtraction of nearly equal vectors.

A reflection does it

$$\mathbf{P} = \mathbf{I} - 2\,\mathbf{v}\mathbf{v}^{T}, \qquad \mathbf{v} = \frac{\mathbf{u}}{|\mathbf{u}|}, \qquad \mathbf{u} = \mathbf{a} - \alpha\,\mathbf{e}_1$$

$\mathbf{P}$ reflects every vector in the mirror perpendicular to $\mathbf{u}$. It is symmetric, orthogonal, and its own inverse: reflecting twice changes nothing.

In the figure, $\mathbf{a} = (3, 4)$ lands on $(-5, 0)$. The arithmetic is on the next slide.



The reflection, by the numbers

Same $\mathbf{a} = (3, 4)$ as in the Gram-Schmidt example. Send it onto the x axis.

the target	$\lvert\mathbf{a}\rvert = 5$; take $\alpha = -5$ (why the minus sign: next slide)
the mirror normal	$\mathbf{u} = \mathbf{a} - \alpha\,\mathbf{e}_1 = (3+5,\ 4) = (8,\ 4), \quad \mathbf{v} = \mathbf{u}/\lvert\mathbf{u}\rvert = (8,\ 4)/\sqrt{80} = (2,\ 1)/\sqrt{5}$
the matrix	$\mathbf{v}\mathbf{v}^T = \frac{1}{5}\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{P} = \mathbf{I} - 2\,\mathbf{v}\mathbf{v}^T = \begin{pmatrix} 1 - \frac{8}{5} & -\frac{4}{5} \\ -\frac{4}{5} & 1 - \frac{2}{5} \end{pmatrix} = \begin{pmatrix} -0.6 & -0.8 \\ -0.8 & 0.6 \end{pmatrix}$
apply it	$\mathbf{P}\mathbf{a} = (-0.6 \cdot 3 - 0.8 \cdot 4, \quad -0.8 \cdot 3 + 0.6 \cdot 4) = (-5,\ 0)$
check	$\mathbf{P}^T\mathbf{P} = \mathbf{I}$ and $\mathbf{P}\mathbf{P} = \mathbf{I}$: orthogonal, and its own inverse

The rows of $\mathbf{P}$ are the Gram-Schmidt directions $\mathbf{e}_1$ and $\mathbf{e}_2$, up to sign. One reflection found both at once.

Which sign for α?

- Both $\alpha = +|\mathbf{a}|$ and $-|\mathbf{a}|$ work on paper: there are two mirrors.
- But $u_1 = a_1 - \alpha$. If $\mathbf{a}$ already points almost along $\mathbf{e}_1$ and α has the same sign as a_1, this subtracts two nearly equal numbers: catastrophic cancellation (Lecture 1).
- $\mathbf{a} = (1, 10^{-9})$: $a_1 - |\mathbf{a}|$ computes to 0 (exact: -5×10^{-19}), and $\mathbf{u}$ points the wrong way. $a_1 + |\mathbf{a}| = 2$ is computed exactly.

Always choose α with the opposite sign to a_1, so that $u_1 = a_1 + \mathrm{sign}(a_1)\,|\mathbf{a}|$ adds.

Column by column

Each reflection works on the rows not yet finished, and leaves the earlier rows and columns alone:

$$\begin{pmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \end{pmatrix} \xrightarrow{\mathbf{P}_1} \begin{pmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \end{pmatrix} \xrightarrow{\mathbf{P}_2} \begin{pmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ 0 & 0 & \times \end{pmatrix} \xrightarrow{\mathbf{P}_3} \begin{pmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ 0 & 0 & 0 \end{pmatrix}$$

So $\mathbf{P}_3\mathbf{P}_2\mathbf{P}_1\mathbf{A} = \mathbf{R}$. Each $\mathbf{P}_i$ is its own inverse, so $\mathbf{A} = (\mathbf{P}_1\mathbf{P}_2\mathbf{P}_3)\,\mathbf{R} = \mathbf{Q}\mathbf{R}$.

Our 2×2 example needs one reflection. Apply the $\mathbf{P}$ we built to both columns: $\mathbf{P}\mathbf{a}_1 = (-5,\ 0)$ and $\mathbf{P}\mathbf{a}_2 = (-0.6 \cdot (-1) - 0.8 \cdot 3,\quad -0.8 \cdot (-1) + 0.6 \cdot 3) = (-1.8,\ 2.6)$, so $\mathbf{R} = \begin{pmatrix} -5 & -1.8 \\ 0 & 2.6 \end{pmatrix}$:

Gram-Schmidt's answer, up to signs.

Why libraries use Householder

- **Accurate:** $\mathbf{Q}$ is orthogonal to rounding precision, for any $\mathbf{A}$.
- **Cheap:** about $\frac{4}{3}n^3$ operations for a square matrix, and $\mathbf{Q}$ is never formed: only the vectors $\mathbf{v}_i$ are stored, and the reflections applied when needed.
- `numpy.linalg.qr` (LAPACK's `geqrf`) and Eigen's `HouseholderQR` both use it.
- Applied from both sides, $\mathbf{PAP}$, the same reflections bring a symmetric matrix to tridiagonal form: the first step of the eigenvalue solvers of Lecture 14.

Part III · The singular value decomposition

Rotate, stretch, rotate

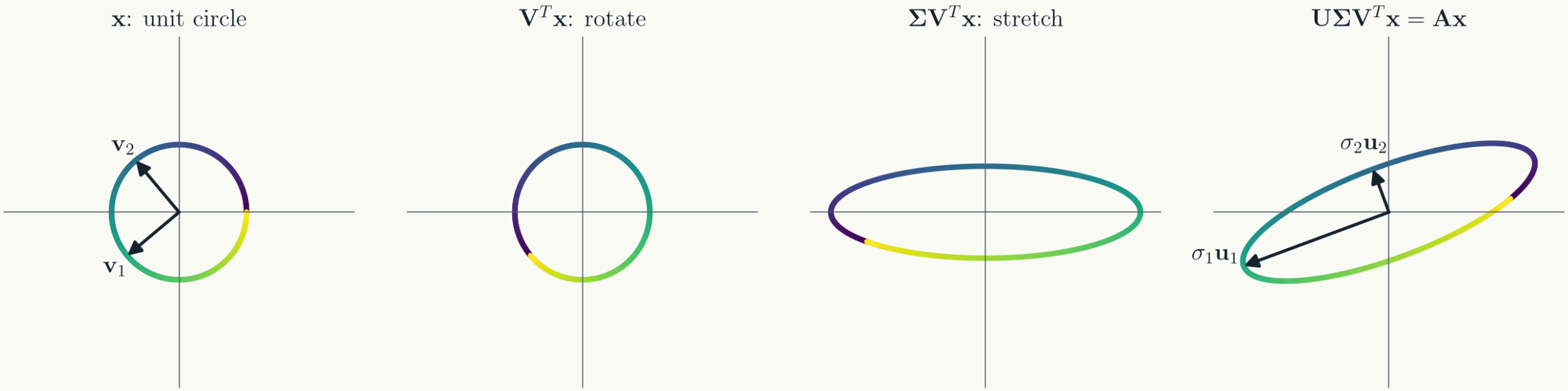
Every matrix has an SVD

Any real $m \times n$ matrix, of any shape and any rank, can be written

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

- $\mathbf{U}$ ($m \times m$) and $\mathbf{V}$ ($n \times n$) are orthogonal.
- $\mathbf{\Sigma}$ ($m \times n$) is diagonal, with the *singular values* $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.
- No conditions at all: unlike LU (square, pivoting), Cholesky (positive definite), or eigenvectors (which may not exist, or may not be orthogonal).

What it means



- $\mathbf{V}^T$ rotates the special input directions $\mathbf{v}_i$ onto the axes; $\mathbf{\Sigma}$ stretches axis i by σ_i; $\mathbf{U}$ rotates the result to the output directions $\mathbf{u}_i$.
- In one line: $\mathbf{A}\mathbf{v}_i = \sigma_i\,\mathbf{u}_i$. The unit circle becomes an ellipse with semi-axes σ_1 and σ_2.

An example

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \end{pmatrix}$$

Singular values $4, 3, \sqrt{5}, 0$, each with a pair $\mathbf{A}\mathbf{v}_i = \sigma_i\,\mathbf{u}_i$. Every row is one multiplication by $\mathbf{A}$:

σ_i	input $\mathbf{v}_i$	$\mathbf{A}\mathbf{v}_i$	output $\mathbf{u}_i$
4	$(0,\ -1,\ 0,\ 0,\ 0)$	$-1\times$ column 2 $= (0,\ 0,\ 0,\ -4)$	$(0,\ 0,\ 0,\ -1)$
3	$(0,\ 0,\ 1,\ 0,\ 0)$	column 3 $= (0,\ 3,\ 0,\ 0)$	$(0,\ 1,\ 0,\ 0)$
$\sqrt{5}$	$(-1,\ 0,\ 0,\ 0,\ -2)/\sqrt{5}$	$(-1-4,\ 0,\ 0,\ 0)/\sqrt{5} = (-\sqrt{5},\ 0,\ 0,\ 0)$	$(-1,\ 0,\ 0,\ 0)$
0	$(-2,\ 0,\ 0,\ 0,\ 1)/\sqrt{5}$	$(-2+2,\ 0,\ 0,\ 0)/\sqrt{5} = \mathbf{0}$	none
0	$(0,\ 0,\ 0,\ 1,\ 0)$	$\mathbf{0}$: column 4 of $\mathbf{A}$ is zero	none

The same thing as matrices

Stack the $\mathbf{u}_i$ as the columns of $\mathbf{U}$, the $\mathbf{v}_i$ as the columns of $\mathbf{V}$, and the σ_i on the diagonal of $\mathbf{\Sigma}$:

$$\mathbf{U} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 0 & 0 & -\frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \end{pmatrix}$$

- $\mathbf{U}$ is 4×4, so it has one more column than there are nonzero σ_i: $(0, 0, 1, 0)$, an output direction nothing maps to. $\mathbf{V}$ is 5×5: its last two columns are the inputs that map to zero.
- Multiply $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ out and $\mathbf{A}$ comes back.

What the SVD reveals

- **Rank:** the number of nonzero singular values. The number of independent directions $\mathbf{A}$ can produce.
- **Range:** all the outputs $\mathbf{Ax}$ can reach. Spanned by the columns of $\mathbf{U}$ with $\sigma_i > 0$. A right-hand side $\mathbf{b}$ outside it has no solution.
- **Nullspace:** all inputs that $\mathbf{A}$ sends to zero. Spanned by the columns of $\mathbf{V}$ with $\sigma_i = 0$ (and any columns beyond m). Adding one to a solution gives another solution.

Rank plus nullspace dimension equals n, the number of unknowns. And the longest and shortest $|\mathbf{Ax}|$ over unit $\mathbf{x}$ are σ_1 and σ_n: the κ of the Gram-Schmidt plot is σ_1 / σ_n, the *condition number*.

Reading the example

- Three nonzero singular values: rank 3.
- Range: the first three columns of $\mathbf{U}$. The fourth, $(0, 0, 1, 0)$, is out of reach: row 3 of $\mathbf{A}$ is zero, so no $\mathbf{x}$ gives $b_3 \neq 0$.
- Nullspace: the last two columns of $\mathbf{V}$, $(-2, 0, 0, 0, 1)/\sqrt{5}$ and $(0, 0, 0, 1, 0)$. Check: $\mathbf{A}\,(-2, 0, 0, 0, 1)^T$ has first entry $-2 + 2 = 0$, and x_4 appears nowhere in $\mathbf{A}$.

Here we can see it by eye. For a 1000×1000 matrix, the SVD finds it for us, and tells us how close to zero each σ_i is.

Summary

- Gram-Schmidt builds $\mathbf{Q}$ by subtracting projections. Clear, but it loses orthogonality when columns are nearly dependent.
- Householder reflections zero a whole column at once and stay orthogonal to rounding precision. That is what libraries use.
- The SVD, $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$: rotate, stretch, rotate. It exists for every matrix and reveals rank, range, and nullspace.

Next time: the SVD at work. Least squares, low-rank approximation, and principal component analysis.